

Ratul Hasan

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Professional Summary

Computer Science graduate with research experience in machine learning, healthcare AI, blockchain, and cybersecurity. Published in peer-reviewed conference proceedings, with research interests in trustworthy AI, large language models, and AI for healthcare. Aspiring to pursue graduate research in Artificial Intelligence.

EXPERIENCE

ELITE Research Lab LLC <i>Research Intern (Remote)</i> - Working on LLM regarding its bias, hallucination - Ongoing journal on LLM’s performance in mental health and depression	Queens, NY, USA May 2026 - Present
BAUET Computer Society, Department of CSE, BAUET <i>Programming Instructor</i> - Conducted regular training sessions on C programming and competitive programming for undergraduate students. - Designed lecture materials, coding exercises, and problem-solving sessions covering algorithms, data structures, and problem-solving techniques. - Mentored students in programming fundamentals and competitive programming, fostering analytical thinking and coding proficiency.	Natore, Bangladesh January 2025 - May 2026
Lab AR <i>Software Engineering Intern</i> - Led a team project to develop a web-based Property Management System. - Contributed to backend development using Laravel (PHP) and collaborated on feature implementation.	Dhaka, Bangladesh July 2025 - August 2025

EDUCATION

Bangladesh Army University of Engineering & Technology (BAUET) <i>B.Sc in Computer Science and Engineering CGPA: 3.82/4.0 Merit Position: 3rd</i> <i>Thesis Title: Biology-Informed Recurrent Neural Networks for Longitudinal Hippocampal Atrophy Forecasting in Alzheimer’s Disease</i>	Natore, Bangladesh 2022 – 2026
Govt. Majid Memorial City College <i>Higher Secondary School Certificate GPA: 5.0/5.0</i>	Khulna, Bangladesh 2019 – 2021

PUBLICATIONS

Conference Publications

Quantum-Resistant FOTA: End-to-End Decentralized Firmware Updates for IoT Using Blockchain and CRYSTALS-Dilithium	December 2025
Ratul Hasan, Md. Momenul Haque, Redoanul Haque, Yousuf Oley, and Ruhani Akter. <i>Proceedings of the 12th International Conference on Next Generation Computing, Communication, Systems and Security (NSysS ’25)</i> . ACM, New York, NY, USA, 110–114. doi.org/10.1145/3777555.3777569	
A Gas-Optimized Blockchain Framework for Scalable Document Verification Using Dynamic Access Control and IPFS	Expected 2026
Ratul Hasan, Samiha Farjana, Yousuf Oley, Md. Ohiduzaman Pranto, and Md Arik Rayhan. Accepted to the <i>International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure (PECCII 2026)</i> . To be published in IEEE Proceedings.	
SplitSafe Ultra: A Moving-Target Defense Framework for Entropy-Aware Encrypted File Persistence on IPFS	Expected 2026
Md. Ahnaf Muhaimin, Ratul Hasan, Yousuf Oley, Samiha Farjana, Bristi Rani Roy, Subrata Kumer Paul, and Md. Ekramul Hamid. Accepted to the <i>5th IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON 2026)</i> . To be published in IEEE Proceedings.	
ActiVisionNet-Fusion: A Dual-Stream Explainable Deep Learning Framework for Driver Drowsiness Detection	Expected 2026
Nasim Ahmed, Ratul Hasan, Subrata Kumer Paul, Dewan Nafiul Islam Noor, Md. Momenul Haque, and Md. Ekramul Hamid. Accepted to the <i>5th IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON 2026)</i> . To be published in IEEE Proceedings.	

Journal Publications

BI-RNN: Biology-Informed Recurrent Neural Networks for Longitudinal Hippocampal Atrophy Forecasting in Alzheimer’s Disease	Under Review
Ratul Hasan, Md. Momenul Haque, Ananya Sarker. Under review in <i>Nature Scientific Reports</i> , Submission Date: May 2026.	

PROJECTS

LedgerSeal: A Blockchain Based Document Verification Framework using Dynamic Access Control and IPFS

[Github](#) 

- Developed a decentralized application (DApp) for constant-time document verification with dynamic access control on IPFS using a public blockchain.
- Tools Used: React JS, Anvil EVM, Foundry, Metamask, Pinata IPFS

BAUET Project & Thesis Archive

[Github](#) 

- Developed a repository website using Django(Both MVT & REST) where theses and projects of BAUET are represented
- Tools Used: HTML, Tailwind CSS, JavaScript, Django, PostgreSQL

ACHIEVEMENTS & CO-CURRICULAR ACTIVITIES

Programming Contests

- 1X ICPC Regionalist (Team: BAUET_Rong_Answer)
- 1X National Collegiate Programming Contest by JU (Rank: 149, Team: BAUET_PARADIGM)
- 2X Inter University Programming Contest (Best Rank: 45 at IUT, Team: BAUET_Twisted_Minds)
- Ranked 10th in Independence Day Programming Contest by Military Institute of Science & Technology
- 1X Champion, 1X 2nd runners-up in Intra University Programming Contest (Team)
- 1X Runners-up in Intra University Programming Contest (Solo)

Others

- 1X 1st Runners Up in Intra University Three Minutes Thesis Presentation

Online Judges

- Codeforces: Max Rating 1650 (Expert)
- Codechef: Max Rating 1676 (3 Star)
- Leetcode: Solved more than 50+ problems
- Solved more than 1000+ problems across various online judges

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TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, SQL || **Frameworks:** Django, REST, Foundry, Laravel

Project Management: Git, Github || **Database:** MySQL, PostgreSQL

Technologies: Machine Learning, Computer Vision, Backend Development, Blockchain

VOLUNTEERING & LEADERSHIP EXPERIENCE

- President, BAUET Computer Society (Advanced through five executive leadership positions)
- Problem Setter, Intra-University Programming Contest, BAUET
- Officer of Documentation, Hult Prize at BAUET

REFERENCES

Prof. Mohammed Golam Sarwar Bhuyan

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